MATH210B Homework 8

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1 Problem 1

Let ζ be a primitive p-th root of unity.

Recall that $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$.

Notice that $\mathbb{Q}(\zeta + \zeta^{-1}) \subseteq \mathbb{R}$ and recall that $[\mathbb{Q}(\zeta) : \mathbb{Q}(\zeta + \zeta^{-1}, \zeta\zeta^{-1})] \leq 2$.

Since $\mathbb{Q}(\zeta)$ is not contained in $\mathbb{R}$, then this is exactly 2.

By the tower law, $[\mathbb{Q}(\zeta + \zeta^{-1}) : \mathbb{Q}] = \frac{p-1}{2}$.

(Notice $\zeta\zeta^{-1} = 1$ so it doesn't contribute anything.)

2 Problem 2

Notice that $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ contains $\sqrt{6}$ as well. Thus, $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3}, \sqrt{2}\sqrt{3})$.

Since the minimal polynomial of $\sqrt{2} + \sqrt{3}$ has degree 4, it's a degree 4 extension.

Notice that $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ is also degree 4. This is because $\sqrt{3}$ is not in $\mathbb{Q}(\sqrt{2})$, which is proven at the end. Thus, these two extensions are equal. However, $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ is separable since it is the splitting field of $(x^2 - 2)(x^2 - 3)$, so we're done.

Lemma 2.1. $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$.

Proof. Assume it is. Then, $\sqrt{3} = a + b\sqrt{2}$ for some $a, b \in Q$. Squaring both sides gives $3 = a^2 + 2ab\sqrt{2} + 2b^2$.

Then, $ab = 0$. Assume first that $a = 0$. Then, $3 = 2b^2$. Assume $b = \frac{p}{q}$ where p, q are coprime. Then, $3q^2 = 2p^2$, so 2 divides q^2, so 2 divides q, so 4 divides q^2, so 2 divides p, which is a contradiction.

Assume now $b = 0$. Then, $a = \sqrt{3}$ which is not true since 3 isn't rational. $\square$

3 Problem 4

Recall that every finite extension of a finite field is normal, since it's a splitting field.

Then, we need to find the smallest d such that 7 divides $5^d - 1$.

This is equivalent to finding the multiplicative order of 5 in $(\mathbb{Z}/7\mathbb{Z})^\times$, which is 6. Thus, $d = 6$.

4 Problem 5

Let $char(F) = p > 0$.

Let E/F be an extension such that $[E : F]$ and p are coprime.

Let $E = F(\alpha_1, \ldots, \alpha_n)$.

Notice that the degree of m_{α_i} is coprime to p since it divides $[E : F]$. Let this degree be n_i.

Then, the derivative of m_{α_i} is non-zero in F since the leading coefficient is monic and so the leading coefficient of the derivative is $n_i \mod p$ which is nonzero. Thus, E is generated by separable elements over F and thus is separable over F.

5 Problem 6

Let E/F be a finite field extension.

Let $\alpha, \beta \in E$ be separable over F.

Then, $F(\alpha, \beta)$ is separable over F since it's generated by separable elements using Corollary 3.5.20 in the notes. But then, $\alpha + \beta$ and $\alpha\beta$ is in E since this is a subfield of E. Notice that this also contains α^{-1}.

Let's now find the maximal separable extension of $F(X)/F(X^4+X^2)$, where F is a field of characteristic 2.

Let's first show that $f(t) = t^4 + t^2 + (X^4 + X^2)$ is the minimal polynomial of X. Clearly, X is a root, so let's show that it's irreducible.

Notice that

$f(t) = t^4 + t^2 + (X^4 + X^2) = (t+X)^4 + (t+X)^2 = (t+X)^4 + (t+X)^2 = (t+X)^2((t+X)^2 + 1)$. But the roots of this are X and $X+1$, so the polynomial doesn't split over $F(X^4+X^2)$. Thus, $F(X)$ is an extension of degree 4. Notice that f isn't separable since it's derivative is zero.

Also notice that X^2 is separable since it's the root of $t(t+1) + (X^4 + X^2)$, which is separable and irreducible using the same argument as above. Thus, the maximal separable subextension is $F(X^2)$.

6 Problem 7

Let $E = F(\alpha_1, \ldots, \alpha_n)$.

By Homework 9 Problem 1 (where I didn't use this problem), we have that the minimal polynomial of every element contains exactly one root. But then the extension should map each α_i to itself, so it acts like the identity on E, and therefore there's exactly one extension.

Now, let's do the backwards direction. Assume there is at most one extension. Then, every minimal polynomial should be of the form $(x - \alpha_1)^m$ for some m since otherwise we would have two extension by mapping one root to another root as in Proposition 3.2.4.

We thus conclude the proof.

7 Problem 8

Recall that in Problem 1 we showed that $\mathbb{Q}[\zeta_p + \zeta_p^{-1}]$ was a degree $\frac{p-1}{2}$. This extension is a subfield of $\mathbb{Q}[\zeta]$. Since this is an Abelian extension, we know that subfields are also Galois extensions. Moreover, we know that n and mth roots of unity, for coprime m, n are linearly disjoint over $\mathbb{Q}$. Then, adding $\zeta_7 + \zeta_7^{-1}$ and $\zeta_{11} + \zeta_{11}^{-1}$ gives us the desired result.

8 Problem 9

Let $f(x) = x^6 + 3$.

Let α be a root of f.

Then, $\alpha^3 = \pm\sqrt{3}i$.

Notice that $\frac{1+\pm\alpha^3}{2}$ is a primitive 6th root of unity.

Thus, the splitting field is generated by α.

Since the action of the Galois group is transitive, α can go to any other root of $x^6 - 3$, but then the Galois group is cyclic of order 6.

9 Problem 10